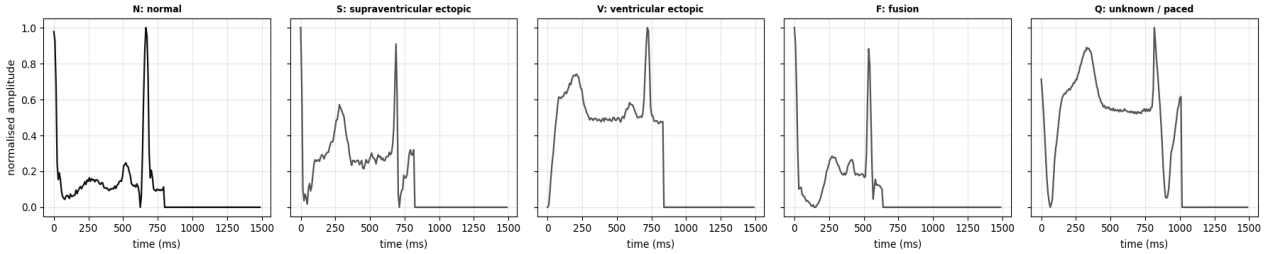
Assignment No. 03: Classifying Heartbeats from ECG Signals

Machine Learning (Fall 2026) — Neural Network Built from Scratch in PyTorch

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1. Data Preparation & Network Architecture

1.1 Example Heartbeats & Waveform Description (Task 1.5)



Normal beats (N) show a clean, narrow QRS peak followed by a smooth return to the baseline, while ventricular ectopic beats (V) originate directly in the ventricles and have a much wider, deeper, and notched waveform. Supraventricular beats (S) have a narrow peak similar to normal beats but fire prematurely, fusion beats (F) look like a mixture of normal and ventricular shapes, and unknown/paced beats (Q) display sharp artificial pacemaker spikes and unusual decay shapes.

1.2 Hand Calculation of Trainable Parameters (Task 2.4)

For our 3-layer network with architecture 187 inputs -> 128 hidden units (ReLU) -> 64 hidden units (ReLU) -> 5 output logits, each layer l has weight matrix W[l] of shape (n[l] x n[l-1]) and bias vector b[l] of shape (n[l]):

Layer	Weight Matrix W[l]	Bias Vector b[l]	Layer Total
Layer 1 (Hidden 1)	128 x 187 = 23,936	128	24,064
Layer 2 (Hidden 2)	64 x 128 = 8,192	64	8,256
Layer 3 (Output)	5 x 64 = 320	5	325
Grand Total	32,448 weights	197 biases	32,645 parameters

1.3 Output-Layer Gradient Check (Task 3.2)

On the first B = 64 training rows, we computed the analytical gradients dJ/dW[3] = (1/B)(Y_hat - Y)^T A[2] and dJ/db[3] = (1/B) sum(Y_hat - Y) by hand and compared them against PyTorch autograd (.grad). The maximum absolute difference was 2.9802e-08 for W[3] and 5.9605e-08 for b[3], which easily passes the < 10^-5 requirement.

2. Complete Results Tables (Section 4)

4.1 Table 1 – Data

Metric	Value
Train / validation / test rows	78,799 / 8,755 / 21,892
Minimum / maximum signal value	0.0000 / 1.0000
Rows per class, train file (N, S, V, F, Q)	72,471, 2,223, 5,788, 641, 6,431 (Total: 87,554)
Rows per class, test file (N, S, V, F, Q)	18,118, 556, 1,448, 162, 1,608 (Total: 21,892)

4.2 Table 2 – Network and training

Metric	Value
Trainable parameters	32,645 (Layer 1: 24,064 | Layer 2: 8,256 | Layer 3: 325)
Initial loss (before training)	1.7202 (Theoretical log 5 = 1.6094)
Gradient check: max difference for W[3] / b[3]	2.98e-08 / 5.96e-08
Chosen alpha	0.5
Best epoch (lowest validation loss)	25
Training / validation loss at best epoch	0.0535 / 0.0716
Training / validation accuracy at best epoch	98.33% / 97.84%
Training time and hardware	36.2s (Colab CPU)

4.3 Table 3 – Learning-rate choice

alpha	Lowest validation loss	Validation accuracy at that epoch	Best epoch
0.01	0.1618	95.79%	30
0.1	0.0867	97.59%	28
0.5	0.0716	97.84%	25

4.4 Table 4 – Test results

Per-class accuracy	N: 99.56%	S: 69.24%	V: 91.78%	F: 54.32%	Q: 97.76%
Overall test metrics	Test accuracy: 97.81%		Balanced accuracy: 82.53%		Always-"N": 82.76%

Full 5 x 5 Test Confusion Matrix (Rows = True Class, Columns = Predicted Class)

True \ Predicted	N (0)	S (1)	V (2)	F (3)	Q (4)	Row Total
N – normal beat (0)	18,039	37	28	1	13	18,118
S – supraventricular (1)	163	385	6	0	2	556
V – ventricular (2)	94	8	1,329	14	3	1,448
F – fusion beat (3)	54	1	19	88	0	162
Q – unknown / paced (4)	25	0	11	0	1,572	1,608

3. Training Curves & Evaluation Figures (Section 4.5)

Figure 2: Validation Loss for Learning Rates {0.01, 0.1, 0.5} (Task 4.2)

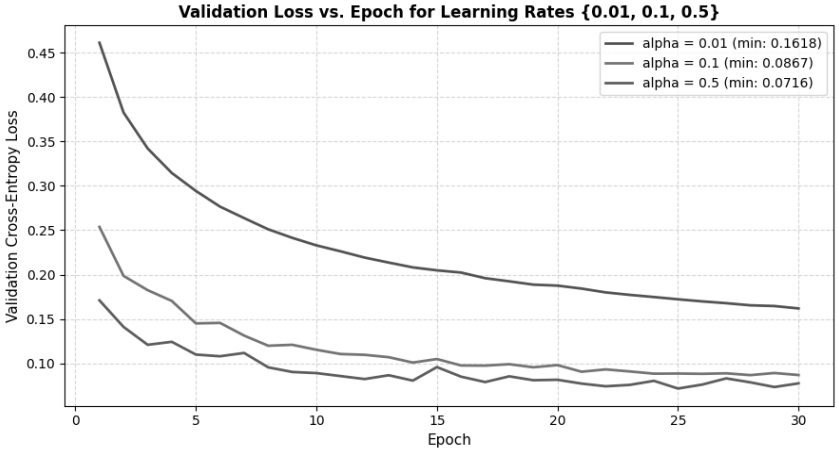


Figure 3: Training and Validation Loss & Accuracy for Chosen alpha = 0.5 (Task 4.3)

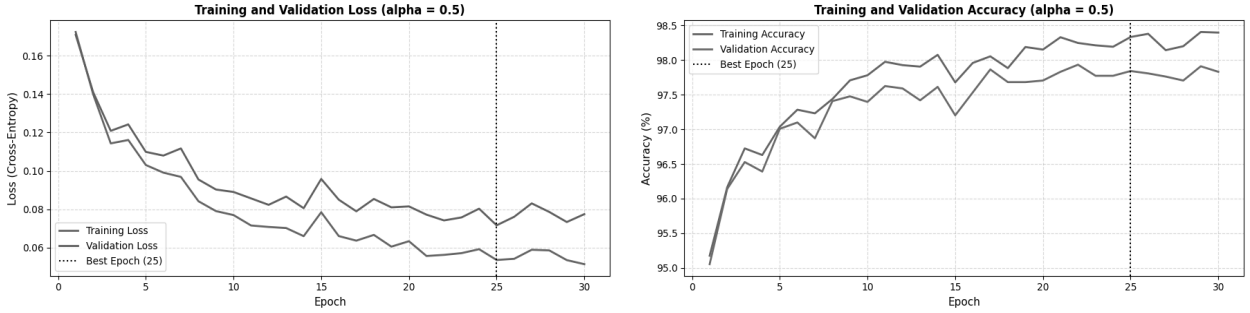
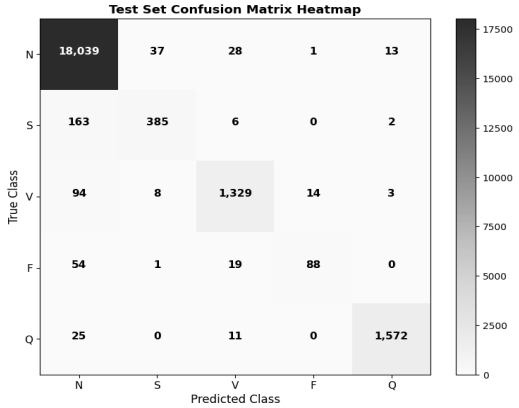
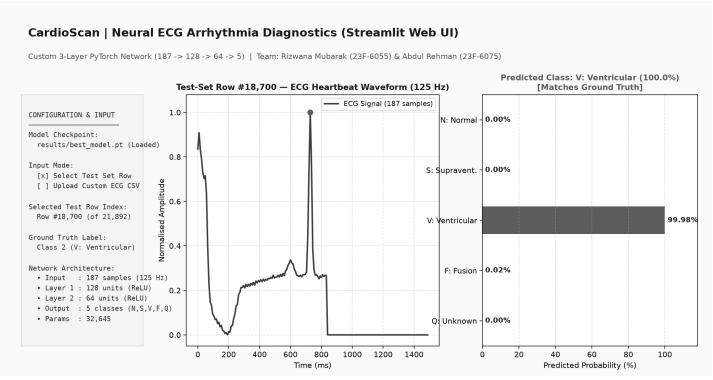


Figure 4 & 5: Test Confusion Matrix Heatmap & Front-End Web UI





4. Task 6 – Analysis Answers

6.1 Always-"N" Baseline Accuracy vs. Our Network

From the class table, the test set has 18,118 normal (N) beats out of 21,892 total rows. A model that always answers "N" would get 18,118 / 21,892 = 82.76% test accuracy. Our trained model achieves 97.81% test accuracy (and 82.53% balanced accuracy), improving on the baseline by 15.05 percentage points. This shows that overall accuracy alone is not a reliable metric when one class dominates the dataset: a useless classifier can score 82.76% accuracy while failing to detect 100% of abnormal heartbeats.

6.2 Lowest Per-Class Accuracy and Confusion Matrix Analysis

Class F (Fusion beat) has the lowest per-class accuracy at 54.32% (88 / 162), followed by Class S at 69.24% (385 / 556). Looking at row F of the confusion matrix, out of 162 true Class F beats, 54 are mistaken for Class N (Normal) and 19 are mistaken for Class V (Ventricular ectopic). These two beat types are hard to separate from Class F because a fusion beat happens when a normal sinus beat and a ventricular ectopic beat depolarize the ventricles at the same time, producing a waveform that is a physical mixture of N and V. In addition, Class F has only 641 training rows (0.73% of the training data).

6.3 Loss Curves and Overfitting Assessment

In our loss curves for alpha = 0.5, both training loss and validation loss drop rapidly during the first 10 epochs and flatten out smoothly through epoch 30. At the best epoch (Epoch 25), the training loss is 0.0535 and the validation loss is 0.0716, leaving a small gap of 0.0181. Because both losses reach low values and the validation loss does not rise while training loss decreases, the network is neither underfitting nor overfitting. The L2 weight decay (lambda = 10^-4) and large dataset size keep generalization strong.

6.4 Why Not Zero Initialization and Why the He Rule Uses sqrt(2 / n[l-1])

If every weight is initialized to zero, all neurons in a hidden layer compute the exact same pre-activation z = 0, produce identical activations, and receive identical gradients during backpropagation. Consequently, all neurons in the layer perform the exact same update and learn the exact same feature (the symmetry problem). For ReLU layers, ReLU(z) = max(z, 0) sets all negative inputs to zero, which cuts the variance of activations in half at each layer. Multiplying the weight variance by 2 (giving standard deviation sqrt(2 / n[l-1])) compensates for the half lost to ReLU and keeps signal variance constant across layers.

6.5 Effect of Momentum on Opposite Gradients and Weight Decay with Zero Gradient

Because the velocity update is v_P = beta * v_P + (1 - beta) * g, when two consecutive mini-batch gradients point in opposite directions, their positive and negative signs partially cancel each other out in v_P. This dampens zig-zagging oscillations across narrow valleys while building speed along consistent descent directions. If a weight receives no loss gradient (dJ/dP = 0), L2 weight decay still contributes g = lambda * P, which shrinks the weight toward zero at each step (P <- P - alpha * v_P).

6.6 Suggested Change to Improve Rare Class Performance (Class F)

To help the network recognize rare classes like Class F (only 641 training rows), we can use Class-Weighted Cross-Entropy Loss. Instead of weighting every sample equally, we multiply the loss of class c by a weight inversely proportional to its training count, w_c = N_total / (5 * N_c), so L = -w_y log softmax(z)_y. This makes misclassifying a rare Class F beat roughly 113 times costlier than misclassifying a Class N beat, forcing the gradients to carve out a clearer decision boundary for rare arrhythmias.